02 Cohort Labels - Cohort Extraction Under Sepsis-3 and Alternative Label Variants

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0.1 Introduction

This notebook applies three Sepsis-3 label variants to the same adult ICU cohort from MIMIC-IV. Following Cohen et al. (2024), the variants use different antibiotic-culture windows and SOFA evaluation periods, producing cohorts that differ in size, event rate, and onset timing.

All three cohorts share the same inclusion criteria (adults $\geq$ 18, first ICU stay, LOS $\geq$ 4 h). The label-variance summary table produced here (M1 deliverable) quantifies these differences before any modelling.

Because all three are Sepsis-3 readings anchored to the same culture-plus-antibiotic trigger, two further variants are built alongside them under Protocol Amendment 2: **E (Anchor-Only)**, the suspicion-of-infection time with the organ-dysfunction requirement removed, and **D (Deterioration)**, acute organ dysfunction with no infection criterion and no treatment timestamp.

Protocol Amendment 4 adds a third, **F (Sepsis-3 Early-Onset)**. Its cohort is built here exactly as Variant B's — same suspicion windows, same SOFA criterion, therefore the same septic stays — and only the onset *timestamp* differs: `03_person_hours.Rmd` moves it back to $\min(t_{\text{susp}}, t_{\text{SOFA}})$, the standard operationalisation of Seymour et al. (2016) and the PhysioNet/CinC 2019 Challenge. All three are reported separately and never enter the pre-registered H1 comparison.

0.2 Base Tables

The three source tables (patients, admissions, and ICU stays) provide demographics, hospital-level outcomes, and the time spine for each ICU episode.

0.3 Base Cohort Criteria

Standard inclusion filters: adults ($\geq$ 18), first ICU stay per patient, and minimum ICU LOS $\geq$ 4 hours.

```
##               step                      label     n n_excluded n_patients
##             <char>                     <char> <int>      <int>      <int>
## 1: icu_stays_total  MIMIC-IV v3.1 ICU stays 94458         NA         NA
## 2:           adult   Age >= 18 at admission 94458          0         NA
## 3:  first_icu_stay First ICU stay per patient 65366      29092         NA
## 4:         min_los         ICU LOS >= 4 hours 65078        288      65078
```

```
## Base cohort: 65,078 stays, 65,078 unique patients
```

0.4 Suspected Infection

Sepsis-3 (Singer et al., 2016) requires temporal co-occurrence of an antibiotic order and a microbiology culture. All antibiotic-culture pairs within each admission are generated here; variant-specific time-window filters are applied in the cohort-building step. Surveillance cultures are excluded.

```
## Antibiotic orders: 820,798
```

```
## Culture samples: 1,530,517
```

```
## ABX-culture pairs (raw): 5,897,222
```

0.5 SOFA Components

The SOFA score (Vincent et al., 1996) quantifies organ dysfunction across six systems. Lab values (platelets, bilirubin, creatinine, PaO_2/FiO_2), vitals (MAP, GCS), and vasopressor records are loaded and filtered to the cohort.

```
## Pre-filtering large tables to 65078 cohort hadm_ids...
```

0.6 Vectorised SOFA Delta

Sepsis-3 requires a SOFA increase $\geq$ 2 points relative to baseline. `data.table::foverlaps` performs interval joins to aggregate measurements within evaluation and baseline windows without per-row loops.

0.7 Build Cohort Per Variant

Each variant applies its own antibiotic-culture time window and SOFA evaluation period, producing a distinct set of labelled onsets. Onsets outside the ICU stay or within the first hour are excluded.

```
## Variant A: 65078 total stays, 6252 sepsis onsets (9.6%)
## Variant B: 65078 total stays, 6447 sepsis onsets (9.9%)
## Variant C: 65078 total stays, 12581 sepsis onsets (19.3%)
## Variant D: 65078 total stays, onsets derived in stage 03
## Variant E: 65078 total stays, 18882 sepsis onsets (29.0%)
## Variant F: 65078 total stays, 6447 sepsis onsets (9.9%)
```

0.8 Label-Variance Table (M1 Deliverable)

The label-variance table summarises cohort size, event count, event rate, and onset-time distribution for each variant. This is the empirical foundation for the label-dominance hypothesis (H1).

```
##
## Saved: 02_label_variance_table.parquet
```

0.9 Alternative-Label Variance Table (Protocol Amendment 2)

Variants A–C are three readings of one definition and share a single culture-plus-antibiotic anchor, so part of their disagreement is definitional rather than empirical. Variant E keeps that anchor and drops the organ-dysfunction requirement; Variant D keeps organ dysfunction and drops the anchor entirely. Variant D's onsets are assigned per hour in `03_person_hours.Rmd` and so are not counted here.

```
##
## Saved: 02_alt_label_variance_table.parquet
```